

38. (b) Apply hit and trial method from the given option. As, here when $a = 11$, $b = 2$, then

$$3 \frac{7}{a} \times b \frac{3}{15} = 3 \frac{7}{11} \times 2 \frac{3}{15} \\ = \frac{40}{11} \times \frac{33}{15} = 8$$

39. (d) Since, $a < b \Rightarrow a - b < 0$. Also, $c < 0$
 $\therefore (a - b)c > 0 \Rightarrow ac - bc > 0 \Rightarrow ac > bc$

40. (c) When we divide a positive integer by another positive integer, the resultant will be a rational number i.e. in the form of p/q , where p and q are positive integers and $q \neq 0$.

41. (d) $2^{12n} - 6^{4n} = (2^{12})^n - (6^4)^n$
 $= (4096)^n - (1296)^n$
 $\therefore 2^{12n} - 6^{4n} = (4096 - 1296)k$
 As $x^n - y^m$ is always divisible by $(x - m)$
 $= 2800(k)$

Hence, last two digits are always be zero.

42. (c) We know that
 $4 = 3p + 1$, for $p = 1$, $9 = 3p$, for $p = 3$
 $16 = 3p + 1$, for $p = 5$,
 $25 = 3p + 1$, for $p = 8$
 $36 = 3p$, for $p = 12$
 Hence, every square integer is of form either $3p$ or $3p + 1$.

43. (d) Given, $r + s = 1$
 For maximum product, $r = s = \frac{1}{2}$
 $\therefore rs = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

44. (a) Required number
 $= [\text{LCM of } 2, 3 \text{ and } 5] + 1$
 $= 30 + 1 = 31$

45. (d) The largest integer that divides product of any four consecutive integers is 24.
 e.g. 1, 2, 3, 4 are four consecutive integers.
 Multiplication $= 1 \times 2 \times 3 \times 4 = 24$
 which divided by 24.

46. (a) When we divide the number 9238 and 7091 by 113, we get the same remainder 85.

47. (d) $d(d(d(12))) = d(d(6))$
 $[\because \text{positive integer divisor of } 12 = 1, 2, 3, 4, 6, 12]$
 $d(d(6)) = d(4)$
 $[\because \text{positive integer divisor of } 6 = 1, 2, 3, 6]$
 and $d(4) = 3$
 $[\because \text{positive integer divisor of } 4 = 1, 2, 4]$

48. (b) Given, by pythagoras theorem,
 $(3)^2 + (4)^2 = (5)^2$
 Let the sides of a right triangle be 3, 4, 5.

Hence, one of its sides is always divisible by 5.

49. (b) $\therefore 1 \leq y \leq 3 \Rightarrow 3 \leq 3y \leq 9 \dots(i)$
 and $-1 \leq x \leq 3 \Rightarrow -4 \leq 4x \leq 12$
 $\Rightarrow -12 \leq -4x \leq 4 \dots(ii)$
 From Eqs. (i) and (ii), we get
 $-9 \leq 3y - 4x \leq 13$

Maximum value is 13.

50. (a) When $p^{11} + 1$ is divided by $(p - 1)$, then the remainder is
 $(1)^{11} + 1 = 1 + 1 = 2$

51. (d) Let $x = a^2$ and $y = b^2$ for some $a, b \in N$, then $xy = a^2b^2 = (ab)^2$, where $ab \in N$.

So, $xy \in A$.

But $\frac{x}{y} = \frac{a^2}{b^2} = \left(\frac{a}{b}\right)^2 \notin A$ as $\frac{a}{b} \notin N$.

52. (d) Since, a ten-digit number is divisible by 4 as well as by 5, then this number must be divisible by 20.

We known that any number is divisible by 20, if last two digits is divisible by 20. It means unit place will be zero and ten's place may be 0, 2, 4, 6 or 8.

53. (c) As, $x < 0 < y$ [given]
 $\Rightarrow x < 0$ and $y > 0$
 $\therefore \frac{1}{x} < 0$ and $\frac{1}{y} > 0 \therefore \frac{1}{x} < \frac{1}{y}$

54. (d) On division of $(19)^n$ by 20, we get remainder either 19 or 1.
 Since, last digit of $(19)^{100}$ is 1.
 $\therefore$ Remainder of $\frac{(19)^{100}}{20}$ is 1.

55. (b) We know that $(x^n - y^n)$ is divisible by $(x - y)$ for all 'n' and is divisible by $(x + y)$ for even 'n'.
 $\therefore (7^{12} - 4^{12})$ is divisible by $(7 + 4) = 11$ and $(7 - 4) = 3$
 Thus, $(7^{12} - 4^{12})$ is divisible by 33.

56. (c) Here, $2^p + 3^q = 17 \dots(i)$
 $2^{p+2} - 3^{q+1} = 5$
 or $4 \cdot 2^p - 3 \cdot 3^q = 5 \dots(ii)$
 On multiplying Eq. (i), by 3 and adding it with Eq. (ii), we get
 $7 \cdot 2^p = 56$, $2^p = 8 = 2^3 \Rightarrow p = 3$
 Put $p = 3$ in Eq. (i), we get
 $2^3 + 3^q = 17$, $3^q = 17 - 8 = 9 = 3^2$
 $\therefore q = 2$

$$57. (c) \frac{49}{15} = 3 + \frac{1}{x + \frac{1}{y + \frac{1}{z}}}$$

$$\Rightarrow \frac{1}{x + \frac{1}{yz + 1}} = \frac{49}{15} - 3 = \frac{4}{15}$$

$$\Rightarrow \frac{1}{x + \frac{1}{yz + 1}} = \frac{4}{15}$$

$$\Rightarrow \frac{yz + 1}{xyz + x + z} = \frac{4}{15}$$

$$\Rightarrow 15(yz + 1) = 4(xyz + x + z)$$

$$\Rightarrow 15yz + 15 = 4xyz + 4x + 4z$$

$$\Rightarrow 15 - 4x = (4xy + 4 - 15y)z$$

$$z = \frac{15 - 4x}{(4xy + 4 - 15y)}$$

As x, y and z are natural numbers.

Again, let $x = 3$ and $y = 1$

$$\text{Then, } z = \frac{15 - 4(3)}{4(3)(1) + 4 - 15(1)}$$

$$= \frac{15 - 12}{12 + 4 - 15} = \frac{3}{1}$$

$\Rightarrow z = 3$, which is a natural number.

58. (a) Required number is
 $x^2 = 540 \times q = 3 \times 3 \times 3 \times 2 \times 2 \times 5 \times q$
 In order to make x^2 a perfect square, the least number we must have to put is
 $q = 3 \times 5 = 15$
 $\therefore x^2 = 540 \times 15 = 8100$

59. (d) We have, $1 + A + A^2 + A^3 = 40$
 $\Rightarrow (1 + A) + A^2(1 + A) = 40$
 $\Rightarrow (1 + A)(1 + A^2) = 40$
 Only $A = 3$ satisfies the above equation,
 so, $A = 3$.

60. (a) We have, $1080 = 2^3 \times 3^3 \times 5$
 For any perfect square, all the powers of the primes have to be even numbers.
 So, if the factor is of the form

$$2^a \times 3^b \times 5^c$$

The values a can be 0 and 2, b can be 0 and 2 and c can take the value 0.

Totally, there are 4 possibilities.

61. (b) Clearly, both statements satisfies divisibility rule of 11.

62. (a) As, we know that a number $a_1 a_2 a_3 a_4 a_5$ is divisible by 9, if sum of the digits, i.e. $a_1 + a_2 + a_3 + a_4 + a_5$ is divisible by 9. Hence, only statement I is true.

63. (c) Taking $p = 11$
 $p + 2 = 13$ [a prime number]
 I. $11 \times 13 + 1 = 144$ [a square number]
 II. $11 + 13 = 24$ [12 is a divisor of 24]
 Hence, both statements I and II are correct.
64. (c) I. If $x = 15$ and $y = 14$, then $x + y = 15 + 14 = 29$, which is a prime number. So, if x and y are composite, then $x + y$ is not always composite.
 II. If $x = 15$ and $y = 14$, then $x - y = 15 - 14 = 1$ which is neither prime nor composite, hence again $x - y$ is not always composite.
 III. Third condition is satisfied for all measure.
 Hence, only III is correct.
65. (c) I. The product of any three consecutive integers is divisible by 6.
 II. Here, $3k = \{\dots - 6, -3, 0, 3, 6, \dots\}$
 $3k + 1 = \{\dots - 5, -2, 1, 4, 7, \dots\}$
 and $3k + 2 = \{\dots - 4, -1, 2, 5, 8, \dots\}$
 $\therefore \{3k, 3k + 1, 3k + 2\}$
 $= \{\dots - 6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6, \dots\}$
 Hence, it is true.
66. (d) All statements are true.
67. (a) (i) If b/a and c/b , then $a = bx$ and $b = cy$ for $x, y \in N$
 $\Rightarrow a = bx = cy(x) = cxy \Rightarrow c/a$
 (ii) If a/b and a/c , then $b = ax$ and $c = ay$ for $xy \in N$,
 $\Rightarrow (b + c) = ax + ay = a(x + y)$
 So, $a/(b + c)$
 (iii) From (ii), we have $(b - c) = a(x - y)$
 $\Rightarrow a/(b - c)$
68. (d) All are true.
69. (c) Given, $33 \mid 231228$ (7006)

$$\begin{array}{r} 231 \\ 228 \\ \hline 198 \\ 198 \\ \hline 30 \end{array}$$

 Now, $33 - 30 = 3$
 So, on adding 3 to 231228, it will be completely divisible by 33.
70. (b) Given, $N = 143k + 28$
 $\Rightarrow N = 143k + 26 + 2$
 $\Rightarrow N = 13(11k + 2) + 2$
 $\therefore$ When the number is divided by 13 the remainder is 2.
71. (a) 1 is neither prime number nor composite number.

72. (b) Here, $88 = 2 \times 2 \times 2 \times 11 = (2)^3 \times (11)^1$
 $91 = (7)^1 \times (13)^1$
 $96 = 2 \times 2 \times 2 \times 2 \times 3$
 $= (2)^5 \times (3)^1$
 and $99 = 3 \times 3 \times 11 = (3)^2 \times (11)^1$
 So, 91 has least number of divisors.
73. (c) Following are the numbers between -11 and 11 which are multiples of 2 or 3.
 $-10, -9, -8, -6, -4, -3, -2,$
 $0, 2, 3, 4, 6, 8, 9, 10$
 So, the numbers of multiples 2 or 3, between -11 and 11 are 15.
74. (b) I. Given, n is a prime number greater than 5.
 Now, $n^4 - 1 = (n^2 - 1)(n^2 + 1)$
 $= (n - 1)(n + 1)(n^2 + 1)$
 Put $n = 11$,
 $n^4 - 1 = (11 - 1)(11 + 1)(121 + 1)$
 [prime number greater than 5]
 $= 10 \times 12 \times 122$
 $= 14640$ which is not divisible by 2400.
 So, statement I is not true.
 II. Every square number can be of the form $5n$ or $(5n \pm 1)$ or $(5n \pm 4)$.
 So, statement II is true.
75. (a) When N is a natural number, then there is only one possible case that N , $(N + 2)$, $(N + 4)$ are prime numbers.
 When $N = 3$, then N , $(N + 2)$, $(N + 4)$
 $= 3, 5, 7$ all are primes numbers.
76. (c) From options,
 (c) $\frac{32}{(3 + 2)} = \frac{32}{5} = 2$ (remainder)
 and $\frac{32}{(3 \times 2)} = \frac{32}{6} = 2$ (remainder)
 $\therefore$ Remainder are same.
 and difference of quotients $= 6 - 5 = 1$
77. (c) If x is a positive even integer and y is negative odd integer, then x^y is a rational number.
78. (b) We know that, between any two rational numbers, there are an infinite number of rational and irrational numbers.
 Hence, only statement II is correct.
79. (d) If k is a positive integer, then every square integer is of the form $4k$ or $4k + 1$, as every square number is either a multiple of 4 or exceeds multiple of 4 by unity.
80. (c) Since, b is largest square divisor of c .
 So, $c = bx$
 [where, x is not a whole square number]
 Also, a^2 divides c .

So, a^2 will divide bx or a will divide b .
 [since, it cannot divide x as it is not a whole square]

81. (b) Every prime number of the form $3k + 1$ can be represented in the form $6m + 1$ only, when k is even.
82. (c) I. In 7710312401, difference between sum of even place digits and the sum of odd place digits 0. So, it is divisible by 11.
 II. Since, $173 < (14)^2$ and it is not divisible by 2, 3, 5, 7, 11 and 13. So, it is a prime number.
 Hence, both statements I and II are correct.
83. (c) Both the statements given are correct. As 121 is the square of 11. So, to obtain prime numbers less than 121, we reject all the multiples of prime numbers less than 11 i.e. 2, 3, 5 and 7. Similarly, every composite number less than 121 is divisible by a prime number less than 11 i.e. 2, 3, 5 or 7.
84. (a) I. $f(k) = 4k + 3$
 For $k = 1$, $f(1) = 4 \times 1 + 3 = 7$
 For $k = 2$, $f(2) = 4 \times 2 + 3 = 11$
 For $k = 3$, $f(3) = 4 \times 3 + 3 = 15$
 Values of $f(k)$ for $k = 1, 2, \dots$ cannot be expressed as sum of two squares, since $1^2 + 2^2 = 5$, $1^2 + 3^2 = 10$,
 $2^2 + 3^2 = 13$.
 II. $f(k) = 8k + 1$
 For $k = 1$, $f(1) = (8 \times 1) + 1 = 9$
 For $k = 2$, $f(2) = (8 \times 2) + 1 = 17$
 For $k = 3$, $f(3) = (8 \times 3) + 1 = 25$
 For $k = 4$, $f(4) = (8 \times 4) + 1 = 33$
 For $k = 5$, $f(5) = (8 \times 5) + 1 = 41$
 $f(k) = 8k + 1$ is square of an odd integer only for some values of k .
 So, only statement I is correct.
85. (d) To check which option is equal to $26^2 + 97^2$, we take the sum of unit digit's square of both number of the question as well as the answer options. Whichever answer option shows the same result will be the answer.
 Here, in $26^2 + 97^2$, $6^2 = 36$
 and $7^2 = 49$
 So, $36 + 49 = 85$
 For option (a), $7^2 + 3^2 = 49 + 9 = 58$
 For option (b), $4^2 + 3^2 = 16 + 9 = 25$
 For option (c), $2^2 + 1^2 = 4 + 1 = 5$
 For option (d), $9^2 + 2^2 = 81 + 4 = 85$
 Only option (d) satisfies the condition.
86. (a) If n is greater than 1, then $n^2(n^2 - 1)$ is always divisible by 12.
Illustration 1 Put $n = 2$, then

$$n^2(n^2 - 1) = (2)^2(2^2 - 1) = 4 \times 3 = 12$$

Illustration 2 Put $n = 3$, then

$$n^2(n^2 - 1) = (3)^2(3^2 - 1) = 9 \times 8 = 72$$

- 87. (a)** We have, $\frac{17^{23} + 29^{23}}{23} + \frac{23^{23}}{23}$
 $(17^{23} + 29^{23})$ is divisible by
 $(17 + 29)$ i.e. 46

So, it is divisible by 23.

Also, 23^{23} is always divisible by 23.

$\therefore$ Remainder = 0

- 88. (b)** Let $E = (1235 \times 4523 \times 2451)$
 $= (12 \times 102 + 11)(12 \times 376 + 11)$
 $\times (12 \times 204 + 3)$

When we divide E by 12, then

Remainder = Remainder when

$11 \times 11 \times 3$ or 363 is divided by $12 = 3$

- 89. (c)** $\because 360 = 2^3 \times 3^2 \times 5^1$
 $\therefore$ Number of divisors
 $= (3 + 1)(2 + 1)(1 + 1)$
 $= 4 \times 3 \times 2 = 24$

- 90. (a)** Given, three-digit number = $XY5$
 and $\frac{XY5}{X215} = Z$

Here, Z can take values 1, 3, 5, 7 and 9.

But only 9 satisfies it,

$$\therefore \begin{array}{r} 135 \\ \times 9 \\ \hline 1215 \end{array}$$

then $X = 1, Y = 3$ and $Z = 9$

Now, $X + Y + Z = 1 + 3 + 9 = 13$

- 91. (d)** Given number is $763X4Y2$.
 Since, given number is divisible by 9.
 $\therefore 7 + 6 + 3 + X + 4 + Y + 2 = 9k$
 $\Rightarrow 22 + X + Y = 9k$
 It is clear that LHS is divisible by 9, if
 $X + Y = 5, 14$

When sum of X and Y is 5, then possible pairs are (1, 4), (4, 1), (2, 3), (3, 2), (0, 5), (5, 0). When sum of X and Y is 14, then possible pairs are (5, 9), (9, 5), (6, 8), (8, 6) and (7, 7).

Hence, the possible pairs are 11.

- 92. (b)** Given, $a_n = 3 - 4n$
 $\therefore \Sigma a_n = \Sigma(3 - 4n)$
 $= 3n - 4 \frac{n \times (n + 1)}{2}$
 $= 3n - 2n^2 - 2n$
 $= n - 2n^2 = -n(2n - 1)$

- 93. (a)** Let the two-digit numbers less than, 50 which when divided by 4 yield unity as remainder be 13, 17, ..., 49.

Here, first term, $a = 13$, common difference, $d = 4$ and $n = 10$

$\therefore$ Required sum = $\frac{n}{2}[2a + (n - 1)d]$

$$= \frac{10}{2}[2 \times 13 + (10 - 1)4]$$

$$= \frac{10}{2}[26 + 36] = \frac{10 \times 62}{2} = 310$$

- 94. (b)** The prime numbers less than 13 are 2, 3, 5, 7, 11.

Also, $p < q < r < 13$

and $p + q + r$ is a prime number.

Hence, only two possible pairs exist i.e. (3, 5, 11) and (5, 7, 11).

- 95. (a)** Product of unit digits
 $= 1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8 \times 9$
 $\times 1 \times 2 \times 3 \times \dots \times 8 \times 9$
 $= 0$

$\therefore$ Required digit in the unit place is 0.

- 96. (c)** See question 88.

- 97. (d)** See example 8.

- 98. (c)** $N = 35 \times 45 \times 55 \times 60 \times 124 \times 75$
 $= 5 \times 7 \times 5 \times 9 \times 5 \times 11 \times 5 \times 12 \times 124$
 $\times 5^2 \times 3$
 $= 5^6 \times 7 \times 9 \times 11 \times 12 \times 124 \times 3$

Hence, the maximum value of m is 6.

- 99. (b)** Given $a - b = 4$... (i)

On squaring both sides, we get

$$(a - b)^2 = (4)^2 \Rightarrow a^2 + b^2 - 2ab = 16$$

$$\Rightarrow 40 - 2ab = 16 \quad [\because a^2 + b^2 = 40]$$

$$\Rightarrow 2ab = 24 \Rightarrow ab = 12$$

$$\therefore a + b = \sqrt{a^2 + b^2 + 2ab}$$

$$= \sqrt{40 + 2 \times 12} = 8$$

$$\therefore a + b = 8 \quad \dots (ii)$$

On solving Eq. (i) and Eq. (ii), we get
 $a = 6$ and $b = 2$

$$\text{Now, } a^3 + b^6 = 6^3 + 2^6 = 2^3 \times 3^3 + 2^6$$

$$= 2^3(3^3 + 2^3)$$

$$= 8(27 + 8) = 8 \times 35 = 280$$

Hence, the value of $a^3 + b^6$ is 280.

- 100. (c)** If factor of given number is of the form of $p_1 \alpha_1, p_2 \alpha_2, \dots, p_n \alpha_n$, then number of prime factors are

$$(\alpha_1 + 1)(\alpha_2 + 1) \dots (\alpha_n + 1)$$

Hence, the prime factor of n are

$$(x_1 + 1)(x_2 + 1) \text{ and } (x_3 + 1)$$

- 101. (b)** See question 57.

- 102. (a)** Let x be the required number.

$$\therefore 52x - 25x = 324 \Rightarrow 27x = 324$$

$$\Rightarrow x = \frac{324}{27} = 12$$

Hence, the required number is 12.

- 103. (a)** $21x + 48y = 5 \Rightarrow 3(7x + 16y) = 5$

If x, y are integer, then LHS of the above equation is multiple of 3, but the RHS of above equation is not multiple of 3.

$\therefore$ There is no any integral values of x and y exist.

- 104. (c)** Let $x = (n^3 - n)(n - 2)$, where $n > 2$
 Take $n = 3$, we get

$$x = (3^3 - 3)(3 - 2) = (27 - 3)(1) = 24$$

[which is divisible by 6, 12 and 24]

$$\text{Take } n = 4, \text{ we get } x = (4^3 - 4)(4 - 2)$$

$$= (64 - 4) \times 2 = 120$$

[which is again divisible by 6, 12 and 24]

Now, take $n = 5$, we get

$$x = (5^3 - 5)(5 - 2) = (125 - 5) \times 3$$

$$= 120 \times 3 = 360$$

[which is again divisible by 6, 12 and 24]

Hence, 24 is the largest natural number.

- 105. (d)** We have, $(234)^{100} + (234)^{101}$
 $\Rightarrow (234)^{100}(1 + 234) = 235(234)^{100}$

We know that square of any number having 4 at unit place is a number in which 6 at unit place.

Any exponent of a number 6 at unit place is always 6 at unit place.

$$\therefore (235)(\dots 6) = \dots 30$$

Resulting number have 0 at unit place.

- 106. (a)** Let the number be y .

$$\therefore y = 7q + 3 \text{ and } q = 11p + 6$$

$$\therefore y = 7(11p + 6) + 3$$

$$\Rightarrow y = 77p + 45$$

When divided by 11 remainder is 1 and quotient is $7p + 4$ and when the quotient is divided by 7 remainder is 4.

- 107. (c)** Seven digits number $876p37q$ is divisible by 225, if this number is divisible by 9 and 5.

If this number is divisible by 9.

Then, sum of its digits is divisible by 9.

Now, sum of digits

$$= 8 + 7 + 6 + p + 3 + 7 + q$$

$$= 31 + p + q$$

$$\therefore p + q = 5 \text{ or } p + q = 14$$

$\therefore$ Given number is divisible by 225 when $q = 5$

if $q = 5, p = 0$ or 9.

- 108. (a)** We have, when $3x + 2y$ is divided by 5, remainder is 2.

$$\therefore 3x + 2y = 5q + 2 \quad \dots (i)$$

and when $2x + 3y$ is divided by 5, the remainder is 3.

$$\therefore 2x + 3y = 5m + 3 \quad \dots(ii)$$

Subtract Eq. (ii) from Eq.(i), we get

$$x - y = 5(q - m) - 1$$

$$x - y = 5(q - m) - 5 + 4$$

$$x - y = 5(q - m - 1) + 4$$

$\therefore x - y$ is divided by 5 remainder is 4.

- 109. (b)** The sum of first 47 terms of the series

$$\frac{1}{4} + \frac{1}{5} - \frac{1}{6} - \frac{1}{4} - \frac{1}{5} + \frac{1}{6} + \frac{1}{4} + \frac{1}{5} - \frac{1}{6} - \frac{1}{4} - \frac{1}{5} + \frac{1}{6} \dots 47 \text{ term}$$

It is clear that sum of first 6 term is zero.

Similarly, sum of first 42 terms is zero.

Sum of last 5 terms

$$= \frac{1}{4} + \frac{1}{5} - \frac{1}{6} - \frac{1}{4} - \frac{1}{5} = \frac{-1}{6}$$

$$\begin{aligned} \text{110. (c)} \quad & \frac{(243 + 647)^2 + (243 - 647)^2}{243 \times 243 + 647 \times 647} \\ &= \frac{2[(243)^2 + (647)^2]}{(243)^2 + (647)^2} = 2 \\ & [\therefore (a + b)^2 + (a - b)^2 = 2(a^2 + b^2)] \end{aligned}$$

$$\begin{aligned} \text{111. (b)} \quad & \text{We have, } N = 90 \times 42 \times 324 \times 55 \\ &= 3^2 \times 10 \times 3 \times 14 \times 3^4 \times 4 \times 55 \\ &= 3^7 \times 10 \times 14 \times 4 \times 55 \end{aligned}$$

Hence, N is divisible by 3^7 .

So, the maximum value of m is 7 when N is divisible by 3^m .

$$\begin{aligned} \text{112. (d)} \quad & 7^{10} - 5^{10} = (7^5)^2 - (5^5)^2 \\ &= (7^5 + 5^5)(7^5 - 5^5) \\ &= (16807 + 3125)(7^5 - 5^5) \\ &= 19932 \times (7^5 - 5^5) \end{aligned}$$

Since, 19932 is divisible by 11.

Hence, $7^{10} - 5^{10}$ is divisible by 11.

$$\text{113. (a)} \quad \text{We have, } a - b = 3$$

Squaring both sides, we get

$$a^2 + b^2 - 2ab = 9 \quad \dots(i)$$

$$\text{Also, } a^3 - b^3 = 117$$

$$\Rightarrow (a - b)(a^2 + b^2 + ab) = 117$$

$$\Rightarrow a^2 + b^2 + ab = \frac{117}{3} = 39 \quad \dots(ii)$$

On subtracting Eq. (i) from Eq. (ii)

$$3ab = 30 \Rightarrow ab = 10$$

$$\text{Now, } a + b = \sqrt{(a - b)^2 + 4ab}$$

$$= \sqrt{9 + 40} = \sqrt{49} = 7$$

- 114. (a)** If R is the remainder obtained by dividing the polynomial $x^n + m^n$ by $(x - m)$, then $(x^n + m^n - R)$ is divisible by $(x - m)$.

$$\text{Let } f(x) = x^n + m^n - R$$

$\therefore f(x)$ is divisible by $(x - m)$.

$$\therefore f(m) = 0$$

$$\Rightarrow (m)^n + (m)^n - R = 0$$

$$R = 2(m)^n$$

As, m is a non-zero integer and n is a positive integer, then R is a non-zero even integer.

- 115. (c)** See question 57.

- 116. (d)** x is prime and y is composite number.

Since, x is prime number.

$\therefore x$ may be $\{2, 3, 5, 7, 11, \dots\}$ and y may be $\{4, 6, 8, 9, 10, 12, 14, 15, \dots\}$

Here, y may be even or odd number and x is odd number except 2.

$\therefore (y - x)$ can be even number.

Also, xy can be even number.

If y is odd integer and $x \geq 3$.

Then, $0.5(x + y)$ is an even integer.

Hence, statements I, II and III are correct.

- 117. (b)** A real number is collection of all rational and irrational numbers.

So, statement II is false.

- 118. (a)** Given, p and q are non-zero integers.

$$\text{I. } \therefore p + q < p - q$$

$$\Rightarrow q + q < 0$$

$$\Rightarrow 2q < 0$$

$$\Rightarrow q < 0$$

$\therefore q$ must be negative.

Hence, statement I is correct.

$$\text{II. } \therefore p + q > p - q$$

$$\Rightarrow q + q > 0$$

$$\therefore 2q > 0$$

$\therefore q$ must be positive irrespective of p .

Hence, statement II is incorrect.

- 119. (d)** I. $a - b < a - c$

$$\Rightarrow b - c > 0 \quad \dots(i)$$

As, b is negative real number and c is positive real number, then Eq. (i) is not true.

II. If $a < b$, when a and b are negative real numbers and c is a positive real number, then $\frac{a}{c} < \frac{b}{c}$ is always

true for $b > a$.

III. $\frac{1}{b} < \frac{1}{c}$ is always true, as c is a positive real number and b is a negative real number.

- 120. (d)** I. If m and n are distinct natural numbers, then $\frac{m}{n} + \frac{n}{m}$ is integer if and only if $m = n$. Hence statement I is incorrect.

$$\begin{aligned} \text{II. } mn \left(\frac{m}{n} + \frac{n}{m} \right) (m^2 + n^2)^{-1} \\ = mn \left(\frac{m^2 + n^2}{mn} \right) \left(\frac{1}{m^2 + n^2} \right) = 1 \end{aligned}$$

Hence, statement II is correct for all values of m and n .

III. Now, $\frac{mn}{m^2 + n^2}$ is a fraction.

So, statement III is incorrect.

02

SEQUENCE AND SERIES

Usually (1-2) questions have been asked from this chapter. Questions are mostly based on relation between arithmetic geometric and harmonic mean.



SEQUENCE

A set of numbers arranged in a definite order according to some definite rule is called a sequence.

e.g. 2, 4, 6, 8, ... is a sequence.

SERIES

If $a_1, a_2, a_3, \dots, a_n$ is a sequence, then the expression $a_1 + a_2 + a_3 + \dots + a_n$ is called the series.

The series is said to be finite or infinite depending upon the last term is given or not. e.g.

(i) $a_1, a_2, a_3, \dots, a_n$ is a finite sequence and is denoted as $\{a_k\}_{k=1}^n$.

(ii) $a_1, a_2, a_3, a_4, \dots$ is an infinite sequence and is denoted by $\{a_n\}_{n=1}^{\infty}$ or simply $\{a_n\}$.

Here, a_1 is called the first term and in case of finite sequence a_n is called the last term.

e.g. 2, 4, 6, 8, ..., 100 is a finite sequence.

1, 2, 3, 4, ... is an infinite sequence.

PROGRESSION

Sequence following certain patterns are called progressions.

e.g. 2, 3, 4, 5, ... is a progression, here each term is increasing by 1.

Arithmetical Progression (AP)

An arithmetic progression is a sequence in which the difference between any term and its preceding term is constant throughout. The constant ' d ' is called the common difference. The first term of an AP is represented by ' a '.